PROJECT DESIGN PHASE

Incident Lifecycle Automation in ServiceNow

Proposed Solution Template | Problem–Solution Fit Template | Solution Architecture

Project Name	Incident Lifecycle Automation in ServiceNow
Phase	3. Project Design Phase
Date	08 September 2026
Team ID	_____

Deliverables and Evaluation Marks

No.	Deliverable	Maximum Marks
1	Proposed Solution Template	2 Marks
2	Problem–Solution Fit Template	2 Marks
3	Solution Architecture	4 Marks

Scope note. This document describes the proposed design of the Incident Lifecycle Automation solution in ServiceNow. Capabilities are presented as design intent. No implementation results, testing outcomes or user research are claimed in this phase document.

1. Proposed Solution Template

Project Name	Incident Lifecycle Automation in ServiceNow
Document Name	Proposed Solution Template
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Maximum Marks	2 Marks

1.1 Purpose

This proposed solution is designed to give Service Desk Agents and Level 2 support teams one consistent, traceable way to handle incidents in ServiceNow. At present, creation, classification, knowledge search, escalation, change handling, child incidents, resolution and documentation can be treated as separate activities. The proposed solution connects them into a single Incident Management lifecycle so that each incident can be handled, followed and documented from start to finish.

1.2 Proposed Solution

S.No.	Parameter	Description
1	**Problem Statement** *(Problem to be solved)*	Service Desk Agents and Level 2 support teams need a consistent way to create, classify, investigate, escalate, resolve and document incidents. The following challenges can delay resolution and reduce traceability: • Inconsistent incident classification, which can lead to incorrect routing. • Incomplete incident information when the incident is created. • Manual reassignment and escalation between support teams. • Difficulty finding relevant knowledge during troubleshooting. • Disconnected related records, such as changes and child incidents. • Incomplete resolution documentation. • Delays in incident resolution caused by the challenges above.
2	**Idea / Solution Description**	Implement a structured Incident Management lifecycle in ServiceNow. The solution connects the following stages: • **Service and Service Offering setup** to provide service context for incidents. • **Incident creation and classification** with complete details and service/CI context. • **Knowledge Management and Agent Assist** to support troubleshooting. • **Reassignment and Level 2 investigation** for incidents that are not resolved at first line. • **Emergency Change Requests** linked when an urgent change is required. • **Child Incidents** created and linked when required. • **Resolution and closure** with cause and resolution details.

S.No.	Parameter	Description
		• **Knowledge Article creation** from useful resolution information. • **SLA visibility** alongside the incident and related records. • **Final validation and testing** to confirm the connected workflow operates as intended.
3	**Novelty / Uniqueness**	The value of the solution lies in connection rather than in any single feature. Incident, knowledge, change, child-incident, SLA and resolution activities are linked into one traceable lifecycle instead of being handled as isolated tasks. The emphasis on connected records and on validation across the complete workflow helps ensure that context is preserved from incident creation to closure and knowledge reuse.
4	**Social Impact / Customer Satisfaction**	Standardized incident handling can make the support experience faster and more consistent for users. Better classification and routing reduce unnecessary delays. Knowledge-assisted troubleshooting helps agents respond with relevant guidance, and clear resolution documentation improves service quality and user confidence.
5	**Business Model** *(Revenue Model)*	This is an internal enterprise IT service-management solution. It is not designed as a standalone revenue-generating product. Its value to the organization comes from improved operational efficiency, reduced handling effort, faster and better resolution, and reuse of organizational knowledge through Knowledge Articles.
6	**Scalability of the Solution**	The ServiceNow platform can support growth in incidents, users, services, configuration items, knowledge articles and related records. Because the lifecycle is standardized, it can be extended to additional support groups and services without changing the core process. Actual capacity depends on the organization's ServiceNow instance, configuration and governance. The solution does not assume unlimited scalability.

Status note. The table above describes the proposed design. Features are stated as design intent; implementation and testing results will be reported separately when supporting evidence is available.

2. Problem–Solution Fit Template

Project Name	Incident Lifecycle Automation in ServiceNow
Document Name	Problem–Solution Fit Template
Phase	3. Project Design Phase
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Team ID	_____
Maximum Marks	2 Marks

2.1 Purpose of the Canvas

The Problem–Solution Fit Canvas checks whether the proposed solution matches the real problems of the people who will use it. It links customer segments, their tasks, triggers, emotions, existing alternatives, constraints and root causes to the proposed ServiceNow workflow. The canvas below adapts the standard 12-section template to Incident Lifecycle Automation. It is based on the project's problem analysis and requirements. No interviews or user research are claimed.

2.2 Problem–Solution Fit Canvas

No.	Canvas Section	Explanation
1	**CUSTOMER SEGMENTS**	• **Service Desk Agents** – first-line users who receive, create and classify incidents and attempt first-line resolution. • **Level 2 Technicians** – handle incidents that need deeper investigation, and record findings and resolution. • **Service Owners / Support Stakeholders** – need reliable incident, service and SLA information to understand service quality. • **Change Stakeholders** – involved when an incident requires an urgent (emergency) change. • **Project Team** – designs, configures and validates the incident lifecycle workflow.
2	**JOBS-TO-BE-DONE / PROBLEMS**	Across the incident lifecycle, users need to: • create complete incidents; • classify incidents correctly using service and CI context; • find relevant troubleshooting knowledge; • route or escalate incidents that are not resolved; • investigate with full context; • connect emergency changes and child incidents; • document the cause and resolution.

No.	Canvas Section	Explanation
3	**TRIGGERS**	Incident-related activities begin when: • a new user issue is reported; • classification is required; • existing knowledge may help to resolve the issue; • the Service Desk cannot resolve the incident and escalation is needed; • an urgent change is needed; • related or recurring issues appear; • the incident is ready for closure.
4	**EMOTIONS: BEFORE / AFTER**	**Before:** users may feel concerned about incomplete information, incorrect routing, delays and missing context when incidents move between teams. **After:** users are expected to feel greater confidence, clarity and traceability, with more satisfaction in the incident process.
5	**AVAILABLE SOLUTIONS**	Existing or alternative approaches include: • **Manual incident handling** – depends on individual judgement. • **Basic Incident Management forms** – record details but do not by themselves connect knowledge, changes and child incidents. • **Separate knowledge search** – done outside the incident, which takes extra effort. • **Manual reassignment** – relies on manual decisions and handoff. • **Separate change / child-incident tracking** – related records may not be clearly connected. • **Informal resolution notes** – difficult to find and reuse.
6	**CUSTOMER CONSTRAINTS**	• The solution must use the existing ServiceNow environment. • Users have different roles and permissions. • The incident and its related records must remain traceable. • Support activities may involve multiple teams.

No.	Canvas Section	Explanation
7	PAINS / ROOT CAUSES	• **Inconsistent classification** – choices depend on individual judgement when service context is not applied consistently. • **Incomplete information** – required details may not be captured when the incident is created. • **Difficulty finding knowledge** – knowledge is searched separately from the incident. • **Manual escalation** – assignment depends on manual decisions. • **Lack of Level 2 context** – earlier details and work notes may not be passed on clearly. • **Disconnected incident/change relationships** – changes and child incidents are tracked separately. • **Difficulty tracking recurring incidents** – related incidents are not clearly linked. • **Incomplete resolution documentation** – cause and resolution details are not recorded consistently. **Common root cause:** incident activities are handled as separate tasks rather than as one connected lifecycle.
8	BEHAVIOUR	**Service Desk Agents** enter incident details, classify issues, search knowledge, assign or escalate incidents and update records. **Level 2 Technicians** review the incident history and work notes, investigate, create or link changes and child incidents when needed, and document the resolution.
9	GAINS / EXPECTED BEHAVIOUR	• Standardized classification and routing. • Faster, better-informed investigation. • Connected related records. • Clearer Level 2 handoff. • Documented resolution and reusable knowledge. • Better SLA visibility and validation of the workflow.
10	OUR SOLUTION	A connected ServiceNow **Incident Lifecycle Automation** workflow (proposed design) that covers: Service and Service Offering setup → incident creation and classification → Knowledge Management / Agent Assist → reassignment and escalation → Level 2 investigation → Emergency Change Request → Child Incidents → resolution and closure → Knowledge Article creation → SLA visibility → final validation and testing.

No.	Canvas Section	Explanation
11	VALUE / FIT	The solution is intended to address each main pain point at the stage where it occurs: • Inconsistent classification → classification supported by service, Service Offering and CI context. • Difficulty finding knowledge → Knowledge Management / Agent Assist within the incident. • Manual escalation and lack of Level 2 context → structured assignment with incident history and work notes. • Disconnected records → Emergency Change, Child Incident, SLA and related-record links. • Incomplete documentation → recorded cause and resolution details and Knowledge Articles.
12	CHANNELS / TOUCHPOINTS	• ServiceNow Incident Management interface • Incident forms / workspace • Service / Service Offering records • Knowledge Management / Agent Assist • Assignment and Level 2 queues • Change Requests • Child Incidents • SLA and related-record views

2.3 Fit Conclusion

The proposed ServiceNow workflow is a suitable fit for the identified problems. Each pain point, such as inconsistent classification, missing context, manual escalation, disconnected records and incomplete documentation, occurs at a specific stage of the incident lifecycle. The proposed solution addresses these pain points at the same stages, keeps related records connected, and supports one consistent end-to-end lifecycle within the existing ServiceNow environment, user roles and traceability needs. Its effectiveness will be confirmed through the planned validation and testing.

3. Solution Architecture

Project Name	Incident Lifecycle Automation in ServiceNow
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Maximum Marks	4 Marks

3.1 Purpose

Solution architecture shows how the business requirements of Incident Management are supported by ServiceNow platform capabilities. It defines the components of the solution, the responsibility of each component and how information moves between them. This helps the project team design, build and validate the workflow in a consistent way and helps reviewers see that each requirement has a place in the design.

3.2 Solution Architecture Overview

The architecture bridges the Incident Management business workflow and the ServiceNow platform capabilities. The Service Desk Agent starts the incident, Incident Management controls the lifecycle, and supporting capabilities provide service/CI context, knowledge assistance, Level 2 investigation, emergency change, child-incident relationships, SLA visibility, resolution documentation and reusable knowledge. Final validation and testing confirm that the connected workflow operates as intended.

Key architectural flow:

- Service Desk Agent → ServiceNow UI / Incident Workspace → Incident Management Lifecycle.
- Incident Management uses Service / Service Offering and CI information for classification and context.
- Knowledge Management / Agent Assist supports troubleshooting and resolution.
- Unresolved incidents can be assigned or escalated to Level 2 Support.
- Level 2 may create/link an Emergency Change and create Child Incidents when required.
- Incident, SLA and related records remain connected for traceability.
- Resolution and closure information can lead to Knowledge Article creation.
- Final validation, testing and deployment validation confirm the complete lifecycle.

3.3 Solution Architecture Diagram

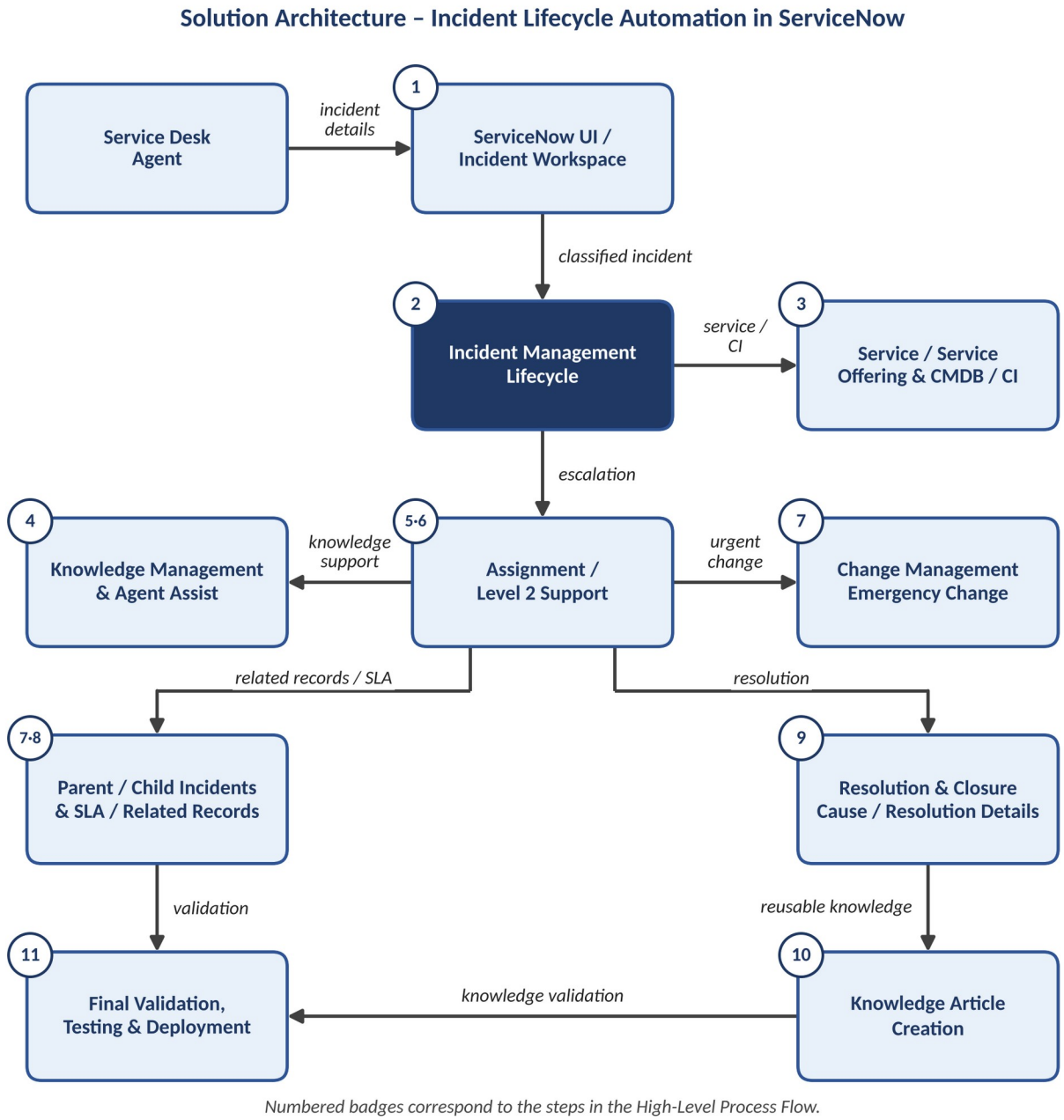


Figure 1: Solution architecture and data flow of the Incident Lifecycle Automation in ServiceNow.

3.4 High-Level Process Flow

The numbered steps below describe the workflow in order. The same numbers appear as badges on the architecture diagram (Figure 1) in Section 3.3.

1. **Access the interface.** The Service Desk Agent accesses the ServiceNow UI / Incident Workspace.
2. **Create the incident.** An incident is created and entered into the Incident Management Lifecycle.
3. **Add context.** Service, Service Offering, CMDB and CI information provide context for classification.

4. **Support troubleshooting.** Knowledge Management / Agent Assist supports troubleshooting and resolution.
5. **Assign or escalate.** Incidents that are not resolved at first line are assigned or escalated to Level 2 Support.
6. **Investigate.** Level 2 investigates and records relevant information, such as findings and work notes.
7. **Link related work.** An Emergency Change Request and Child Incidents are created or linked when required.
8. **Maintain traceability.** SLA and related records remain connected to the incident for traceability.
9. **Document resolution.** Resolution and closure details, including the cause and resolution, are documented.
10. **Reuse knowledge.** Knowledge Articles are created from useful resolution information.
11. **Validate.** Final validation, testing and deployment validation confirm the complete workflow.

Figure 1 also connects knowledge support to the Assignment / Level 2 stage, so relevant knowledge remains available during investigation as well as at first line.

3.5 Component Description

The table lists the components shown in Figure 1. In the diagram, Parent / Child Incidents and SLA / Related Records share one box because they are handled together as connected records.

Component	Purpose	Responsibility	Interaction with other components
Service Desk Agent	First-line user who starts the incident process.	Reports and records the incident and attempts first-line resolution.	Sends incident details to the ServiceNow UI / Incident Workspace.
ServiceNow UI / Incident Workspace	Working interface for incident handling.	Provides the forms and views used to enter, review and update incident information.	Receives incident details from the agent and passes the classified incident to the Incident Management Lifecycle.
Incident Management Lifecycle	Central process that controls the incident from creation to closure.	Controls incident stages and coordinates classification, assignment and escalation.	Uses service/CI context; sends escalations to Assignment / Level 2 Support.
Service / Service Offering / CMDB / CI	Provides context about what is affected.	Identifies the service, Service Offering and configuration item related to the incident.	Supplies service/CI information to Incident Management for classification and context.
Knowledge Management / Agent Assist	Supports troubleshooting with existing knowledge.	Provides relevant knowledge to help diagnose and resolve incidents.	Connected to the incident and to Assignment / Level 2 Support as knowledge support.
Assignment / Level 2 Support	Handles incidents not resolved at first line.	Receives assigned or escalated incidents, investigates and records findings.	Receives escalation from Incident Management; uses knowledge support; raises urgent change; links related records; moves to resolution.

Component	Purpose	Responsibility	Interaction with other components
Change Management / Emergency Change	Handles urgent changes linked to an incident.	Records and tracks an Emergency Change Request when Level 2 identifies an urgent change.	Receives the urgent change request from Level 2 and stays linked to the incident.
Parent / Child Incidents	Keep related incident work connected.	Link Child Incidents to the parent incident when required.	Created from Level 2 investigation; kept with the related records.
SLA and Related Records	Show service-level status and keep records connected.	Keep SLA information and related records visible with the incident.	Linked to the incident and to Level 2 work; provide input for final validation.
Resolution & Closure	Documents how the incident was resolved.	Records the cause and resolution details and closes the incident.	Receives the resolution from Level 2; provides reusable information to Knowledge Article Creation.
Knowledge Article Creation	Turns resolution information into reusable knowledge.	Creates Knowledge Articles from useful resolution information.	Receives reusable knowledge from Resolution & Closure; its result is included in final validation.
Final Validation, Testing & Deployment	Confirms the lifecycle works as intended.	Validates and tests the connected workflow and the deployment validation.	Receives validation input from related records and Knowledge Article Creation.

3.6 Architecture Benefits

- **Consistent incident handling** – every incident follows the same controlled lifecycle.
- **Better traceability** – the incident, service, change, child-incident and SLA records remain connected.
- **Improved investigation** – Level 2 receives the service/CI context, history and knowledge support.
- **Connected records** – emergency changes and child incidents are linked rather than tracked separately.
- **Knowledge reuse** – resolution information can become Knowledge Articles for future incidents.
- **Validation of the incident lifecycle** – final validation and testing confirm that the connected workflow operates as intended.

Status note. The architecture describes the proposed design. Implementation and testing evidence will be documented separately.